

ISLAMABAD INTL, Pakistan

WMO: 415710

Lat: 33.617N

Lon: 73.099E

Elev: 508

StdP: 95.36

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.3	(c) 3.6	(d) -3.1	(e) 3.1	(f) 16.4	(g) -1.2	(h) 3.6	(i) 12.3	(j) 11.8	(k) 14.2	(l) 10.3	(m) 15.4	(n) 0.0	(o) 230	(p) 0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 12.9	(c) 41.1	(d) 22.8	(e) 39.2	(f) 23.0	(g) 38.0	(h) 22.8	(i) 28.1	(j) 34.3	(k) 27.5	(l) 33.6	(m) 27.0	(n) 32.8	(o) 5.9	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.5	23.5	31.5	26.0	22.8	31.2	25.4	21.9	30.5	94.2	34.1	91.4	34.0	88.7	32.9	31.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.0	(b) 10.4	(c) 9.3	(e) 0.4	(f) 43.7	(g) 0.9	(h) 1.7	(i) -0.3	(j) 44.9	(k) -0.8	(l) 45.9	(m) -1.3	(n) 46.9	(o) -2.0	(p) 48.1		
(a) 13.0	(b) 10.4	(c) 9.3	(e) -0.3	(f) 29.5	(g) 1.1	(h) 1.0	(i) -1.1	(j) 30.2	(k) -1.7	(l) 30.8	(m) -2.3	(n) 31.3	(o) -3.1	(p) 32.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.4	11.1	13.8	18.9	24.0	29.0	31.4	30.1	29.0	27.6	23.4	17.2	12.6		
	DBStd	7.51	2.04	2.59	3.21	3.44	3.16	2.66	2.46	1.96	2.02	2.71	2.30	2.02		
	HDD10.0	16	12	2	0	0	0	0	0	0	0	0	0	2		
	HDD18.3	613	225	128	31	3	0	0	0	0	0	1	48	177		
	CDD10.0	4539	44	108	277	419	590	643	623	589	529	417	217	84		
	CDD18.3	2095	0	1	51	172	332	393	364	331	279	159	14	0		
	CDH23.3	27184	1	24	551	1952	4637	5943	5022	4151	3119	1554	222	8		
	CDH26.7	14823	0	1	167	970	2833	3818	2797	2043	1513	656	26	0		
(14) Wind	WSAvg	3.0	1.9	2.7	3.3	3.9	4.2	4.5	4.0	3.3	2.7	2.1	1.5	1.4		
(15) Precipitation	PrecAvg	1054	61	73	74	50	34	65	240	285	107	25	17	29		
	PrecMax	1596	147	208	172	210	115	239	618	497	279	96	70	152		
	PrecMin	709	0	0	0	6	3	7	41	82	14	0	0	0		
	PrecStd	227	46	51	44	44	25	60	132	118	67	26	19	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	26.1	32.8	38.4	42.0	43.9	41.0	37.1	36.1	34.8	28.9	24.2		
		MCWB	11.8	15.0	18.3	20.1	21.5	22.5	25.7	27.1	25.3	23.3	17.2	13.5		
	2%	DB	20.4	23.8	30.8	36.1	40.4	41.9	38.9	35.9	35.1	33.1	27.2	22.2		
		MCWB	11.3	13.4	17.5	19.5	21.4	23.0	25.9	27.0	25.1	21.5	16.6	12.7		
	5%	DB	18.9	22.0	28.2	34.1	38.8	40.2	37.1	35.0	34.1	32.0	26.0	20.9		
		MCWB	10.7	12.6	16.4	19.0	21.3	23.0	26.0	26.6	24.6	21.0	16.0	12.1		
	10%	DB	17.1	20.3	26.4	32.2	37.1	38.9	35.8	33.9	33.1	30.8	24.2	19.1		
		MCWB	10.1	12.2	15.7	18.5	20.8	23.3	26.0	26.3	24.2	20.2	15.0	11.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.5	16.5	20.0	21.7	24.7	27.7	28.9	28.6	27.8	25.0	19.2	15.1		
		MCDB	19.0	23.6	30.1	33.6	37.5	37.3	35.0	34.4	33.0	32.1	26.3	22.0		
	2%	WB	12.4	14.8	18.5	20.7	23.4	26.8	28.1	28.1	26.6	23.6	17.8	13.8		
		MCDB	17.9	20.9	27.9	32.5	36.6	36.4	34.6	33.7	32.7	30.4	25.3	20.1		
	5%	WB	11.6	13.8	17.5	20.0	22.6	26.0	27.6	27.5	25.9	22.4	16.8	12.8		
		MCDB	17.0	19.5	25.9	31.4	35.5	35.0	33.9	32.9	32.0	29.3	23.9	19.1		
	10%	WB	10.7	12.9	16.6	19.2	21.7	25.2	27.0	27.0	25.3	21.2	15.7	12.0		
		MCDB	15.9	18.8	24.3	30.0	34.2	34.2	33.0	32.0	31.1	28.7	22.6	17.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	11.8	12.8	13.5	13.9	12.9	9.5	8.9	10.9	13.9	15.0	14.1		
		MCDBR	15.7	14.9	15.6	16.3	16.1	15.4	11.6	10.2	11.7	14.4	16.4	16.6		
	5% WB	MCWBR	7.8	6.7	6.0	5.5	5.3	5.1	4.2	4.1	4.5	5.5	7.3	8.2		
		MCWBR	12.8	12.3	13.4	14.0	14.3	11.5	9.2	9.0	10.1	12.3	14.7	14.3		
(40) Clear-Sky Solar Irradiance	taub		0.426	0.479	0.486	0.509	0.539	0.624	0.696	0.661	0.565	0.520	0.476	0.428		
	taud		1.999	1.915	1.934	1.923	1.868	1.720	1.636	1.703	1.857	1.859	1.901	2.005		
	Ebn at Noon		747	750	787	791	770	704	651	664	709	699	684	716		
	Edh at Noon		141	170	180	190	203	235	254	234	192	177	154	133		
(44) All-Sky Solar Radiation	RadAvg		2.92	3.48	4.81	6.01	7.02	7.03	5.99	5.45	5.36	4.67	3.47	2.86		
	RadStd		0.32	0.48	0.49	0.36	0.34	0.33	0.45	0.42	0.33	0.25	0.20	0.19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		+0.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page